CS 4510-X

Parikh's Theorem

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1.1 Deliverables

There are 9 problems on this sheet (five during the proof and four at the end). Turn in 6 of the 9.

1.2 Introduction

Parikh's theorem states that context-free languages are equivalent to regular languages if you ignore the order of the letters in a word. We will prove the theorem together by working through several flawed proofs, and then explore some consequences of the theorem.

Definition 1.1 *Given an alphabet Σ with characters $s_1, ..., s_{|\Sigma|}$, define the function $\Psi : \Sigma^* \to \mathbb{N}^{|\Sigma|}$ by $\Psi(w) = [n_1, ..., n_{|\Sigma|}]$ where n_i is the number of occurrences of s_i in w. As an example, let $\Sigma = \{a, b, c\}$. Then $\Psi(ababa) = [3, 2, 0]$.*

Definition 1.2 *For a language L, define $\Psi(L) = \{\Psi(w) \mid w \in L\}$*

Theorem 1.3 (Parikh's Theorem) *For any context-free language L, there is a regular language R such that $\Psi(L) = \Psi(R)$. We say that L and R are "letter-equivalent".*

Problem 1: Let $L_1 = \{a^n b^n \mid n \in \mathbb{N}\}$ and let $L_2 = \{w \mid w \text{ is a palindrome}\}$. Give regular languages R_1 and R_2 such that $\Psi(L_1) = \Psi(R_1)$ and $\Psi(L_2) = \Psi(R_2)$.

Proof: Let R_1 be defined by the regular expression $(ab)^*$. First we show that $\Psi(L_1) \subseteq \Psi(R_1)$. Choosing any $x \in \Psi(L_1)$ so this x must be some set $[i_a, i_b]$ which corresponds to some $a^n b^n \in L_1$. Specifically x has to be of the form $[n, n]$. Now consider the string $\underbrace{ab \ldots ab}_{\text{ab } n \text{ times}}$. This string consists of n a's and n b's and hence curresponds to $[n, n]$. Hence we have $x \in \Psi(R_1)$ as well. Now similarly we now show that $\Psi(R_1) \subseteq \Psi(L_1)$ consider some $x \in \Psi(R_1)$ corresponding to some $[i_a, i_b]$, note again in this case as well because all strings are of the form $(ab)^*$, the count of a and b are equal and hence it's all of the form $[n, n]$. But for any of this form we can find a corresponding $w \in L_1$ defined as $a^n b^n$ such that $\Psi(w) = [n, n]$ and hence for any $x \in \Psi R_1$ we have $x \in \Psi(L_1)$ because for any word in R_1 we can find a corresponding word in L_1 that have the same Ψ and vice versa. Hence we have $\Psi(R_1) = \Psi(L_1)$

Let R_2 be defined by $\{(a \cup \epsilon)(aa)^*(bb)^* \cup (b \cup \epsilon)(aa)^*(bb)^*\}$. Note that this string is essentially a string with an even count of each letter OR an even count of individual letters plus an additional single letter. Now take some $x \in \Psi(L_1)$, so we have some $[i_a, i_b]$ such that either both are even or only one of them is odd. Now assume both are even. Now let them be $[2m, 2n]$. Using the construct $a^{2m} b^{2n}$ and note that this string is in the regular language that we defined and hence we have $x \in \Psi(R_1)$ and WLOG if we have $[2m+1, 2n]$ then consider $aa^{2m}b^{2n}$ and this is in our regular language as defined by the regex above. So we have $\Psi(L_1) \subseteq \Psi(R_1)$. Now consider the other direction and take some $x \in \Psi(R_1)$ and note that we have again we have the same $[i_a, i_b]$ as above. So assume again we have $[2m, 2n]$ in this case just consider $a^m b^n b^n a^m$, clearly this is a palindrome with $[2m, 2n]$ and hence we have $x \in \Psi(L_1)$ in this case. Consider WLOG the case where we have $[2m+1, 2n]$. In this case consider the string $a^m b^n a b^n a^m$, clearly again we hvae $\Psi(a^m b^n a b^n a^m) = [2m+1, 2n]$ and hence we have $x \in \Psi(L_1)$. This gives us $\Psi(R_1) \subseteq \Psi(L_1)$ which gives us $\Psi(R_1) = \Psi(L_1)$ ■

How do we prove Parikh's theorem? The core idea of the proof relies on the concept from the pumping lemma of pumping downwards. Making a shorter string from a longer string allows us to do strong induction on the length of the string. Furthermore, there are a finite number of ways to pump a string downwards, and finite sets play nice with regular languages.scope.com/

Definition 1.4 *Given a context-free language L, let G be a grammar that generates L and let p be the pumping length of L. Define $B = \{w \in L \mid |w| < p\}$ and $C = \{xy \in \Sigma^* \mid |xy| \leq p$ and for some nonterminal A in G, $A \overset{*}{\Rightarrow} xAy\}$.*

Dubious Claim 1.5 *Clearly BC^* is a regular language. Perhaps we could show that $\Psi(L) = \Psi(BC^*)$.*

Problem 2: Show that $\Psi(L) \subseteq \Psi(BC^*)$, using the pumping lemma and strong induction on the length of the string.

Proof: First note that for any $w \in L$ for which we have $|w| < p$ clearly we have $w \in B$ and hence $w \in BC^*$ so this automatically gives us $\Psi(w) \in \Psi(L)$ and $\Psi(w) \in \Psi(BC^*)$. Now we'll do induction, the base case is for $|w|$ from 1 to p which is clearly true as we showed above. Now let's assume true for some arbitrary string of length $n-p, \ldots, n$ (from strong induction). Now we need to show that the case for $n+1$ is true. Now if the case for $n+1$ is true first we write $w = uvxyz$ (as per the pumping lemma), now note that we can find this such that $|vxy| \leq p$. Now rearrange the string to $uzvxy$, first note that uzx is in L ($i = 0$ in pumping lemma) and because of strong induction base cases (note by removing vy we're removing at most p elements, so our assumption step i.e the last p steps are true covers this case) we have some rearrangement of this is in BC^* as well so WLOG let this be bc such that $|b| < p$ and $c \in C^*$, now consider $uzvxy$ consider the rearrangement $uzxvy$ and as $uzx = bc$ we have this as $bcvy$ such that $|b| < p, c \in C^*$ and vy are from the pumping lemma. It is enough now to show that $vy \in C$ because concat is closed in C^*. Now as per the pumping lemma we can pump $v^i xy^i$ for any arbitrary i, this necessarily implies that there is a production such that $A \to^* vAy$ as we need to keep the center constant and v, y increase together. So this means that $vy \in C^*$ and hence $cvy \in C^*$. Let $cvy = k \in C^*$, so we just wrote the rearrangement as bc' where $|b| < p$ and $c' \in C^*$ and hence this implies that we have $\Psi(w) \in \Psi(BC^*)$. ■

Now we try to prove $\Psi(BC^n) \subseteq \Psi(L)$ for all n by induction. Since $B \subseteq L$, $\Psi(B) \subseteq \Psi(L)$. Now, suppose $\Psi(BC^i) \subseteq \Psi(L)$. If $w \in BC^{i+1}$, we can write $w = w_0 s$, where $w_0 \in BC^i$ and $s \in C$. By the inductive hypothesis, there is a word $w' \in L$ with $\Psi(w') = \Psi(w_0)$. We know that there is some nonterminal A such that $s = xy$ and $A \overset{*}{\Rightarrow} xAy$. Unfortunately, there is no way to complete this step, since the derivation of w doesn't necessarily contain A.[1]

[1] Of course, failing to prove a statement doesn't mean that the statement is false. Can you give an example of a grammar G such that $\Psi(BC^*) \not\subseteq \Psi(L(G))$?

So what do we do? Well, the inductive proof above would have worked out if we had a way of forcing the derivation of w to contain the terminals we need. This gives rise to the following idea: separate the words of L into subsets as determined by the nonterminals used in their derivations. Then, if we can show that each subset is letter-equivalent to a regular language, we will be done. (Why?)

Definition 1.6 *For any U, a subset of the nonterminals of G, define $L_U = \{w \mid w \in L$ and some derivation of w by G contains every nonterminal in $U\}$. (Note that $\cup_U L_U = L$.) Then define $B_U = \{w \in L_U \mid |w| < p\}$ and $C_U = \{xy \in \Sigma^* \mid |xy| \leq p$ and for some nonterminal A in U, $A \overset{*}{\Rightarrow} xAy\}$.*

Dubious Claim 1.7 $\Psi(L_U) = \Psi(B_U C_U^*)$

Problem 3: Following the proof outline at the bottom of page 1, show that $\Psi(B_U C_U^n) \subseteq \Psi(L_U)$ for $n \in \mathbb{N}$.

Proof: First consider $B_U C_U^1$ and take a string of the form wxy such that $w \in B_U$ and $xy \in C_U$, now let A be the non-terminal such that in C_U we have $A \to^* xAy$, now as $w \in L_U$, the derivations of w contains every non-terminal which means it also contains A. So say we have $S \to^* ...A... \to^* w$, but note we have $A \to^* xAy$ this means that we can have $S \to^* ...A... \to^* ...xAy...$ and now let if we do the same productions as we did to generate w, we essentially generate a string such that it contains the substrings w, x and y. Hence the string we generate say w' is a rearrangement of wxy and therefore $\Psi(wxy) = \Psi(w')$ and $wxy \in B_U C_U$ and $w' \in L_U$. Now assume true for some arbitrary i so we have $B_U C_U^i$ and a string $wx_1y_1x_2y_2 \dots x_iy_i$ (repetitions allowed) such that $w \in B_U$ and $x_jy_j \in C_U$ for $1 \leq j \leq i$. Now we need to show the case for $i+1$. So we have $wx_1y_1x_2y_2 \dots x_iy_i$, now consider the $i+1$ so say we have, $wx_1y_1x_2y_2 \dots x_iy_ix_{i+1}y_{i+1}$. Now clearly for $wx_1y_1x_2y_2 \dots x_iy_i$ because of the inductive assumption, some rearrangement of this is in L_U so let that rearrangement by r now as $r \in L_U$ we have some $S \to^* r$ that contains all the productions in L_U. But note now that considering $x_{i+1}y_{i+1}$ this is in C_U and hence for some A we have $A \to^* x_{i+1}y_{i+1}$ so we must have this non terminal A in the production of r so we have $S \to^* ...A... \to^* r$, doing the same as above we do $A \to^* x_{i+1}y_{i+1}$ and hence introduce the string r' which contains r, x_{i+1}, y_{i+1} or in other words is one rearrangement of $rx_{i+1}y_{i+1}$ and hence that implies $\Psi(rx_{i+1}y_{i+1}) = \Psi(r')$ and consequently we have $\Psi(wx_1y_1x_2y_2 \dots x_iy_ix_{i+1}y_{i+1}) \in \Psi(L_U)$ implying that $\Psi(B_U C_U^{i+1}) \subseteq \Psi(L_U)$, and this completes our inductive step. This shows that for any $n \in$ we have $\Psi(B_U C_U^n) \subseteq \Psi(L_U)$. ■

Unfortunately, now the inductive proof that you gave above for $\Psi(L) \subseteq \Psi(BC^*)$ doesn't work to show $\Psi(L_U) \subseteq \Psi(B_U C_U^*)$.

Problem 4: Why not? (Hint: pumping down doesn't quite work anymore, as we're no longer inducting over strings of L!)

Proof: The reason why it doesn't work anymore is because our the inductive assumption for the cases 1 to n don't work as when we consider substrings of length n we are guaranteed strings in L and not in L_U. And as in this case we're considering the set L_U and so as we can't guarantee that the strings are not in L_U we cannot use any of our inductive hypothesis. ■

We're almost there! But we need a slightly stronger version of the pumping lemma.

Theorem 1.8 (Stronger Pumping Lemma for Context-Free Languages) *Let L be a context-free language. Then there exists a p such that, if $w \in L$ and $|w| \geq p^k$, then there is some ter-*

minal A such that $S \overset{*}{\Rightarrow} uAz \overset{*}{\Rightarrow} uv_1Ay_1z \overset{*}{\Rightarrow} uv_1v_2Ay_2y_1z \overset{*}{\Rightarrow} ... \overset{*}{\Rightarrow} uv_1v_2...v_kAy_k...y_2y_1z \overset{*}{\Rightarrow} uv_1v_2...v_kxy_k...y_2y_1z = w$, with $|v_1v_2...v_kxy_k...y_2y_1| \leq p^k$. Note that for $k = 1$ this is just the ordinary pumping lemma.

The proof of this stronger form of the pumping lemma follows the same structure as the proof of the normal pumping lemma: if a derivation tree is tall enough, it must contain a long path, and a long enough path must contain some nonterminal repeated at least $k + 1$ times.

Now, let $k = |U|$ and after the following definitions we can finally prove Parikh's theorem.

Definition 1.9 *Let $B'_U = \{w \in L_U \mid |w| < p^k\}$ and $C'_U = \{xy \in \Sigma^* \mid |xy| \leq p^k$ and for some nonterminal A in U, $A \overset{*}{\Rightarrow} xAy\}$.*

Correct Claim 1.10 $\Psi(L_U) = \Psi(B'_U C'^*_U)$

<u>Problem 5:</u> $\Psi(B'_U C'^*_U) \subseteq \Psi(L_U)$ follows almost unchanged from the proof you gave in problem 3. Now, using the Stronger Pumping Lemma for CFL's, prove the reverse direction, i.e. that $\Psi(L_U) \subseteq \Psi(B'_U C'^*_U)$. (The proof closely follows the original proof in problem 2, but now you can pump down in a certain way to stay in L_U.)

Proof: First note that for any $w \in L_U$ clearly if $|w| < p^k$ clearly it's in $B'_U {C'_U}^0 = B'_U$. So take $|w| > p^k$. Now if this is the case using the stronger pumping lemma we have a terminal A such that $S \to^* uAz \to^* uv_1Ay_1z \to^* uv_1v_2Ay_2y_1z \to^* uv_1 \ldots v_kAy_k \ldots y_1z \to^* uv_1 \ldots v_kxy_k \ldots y_1z = w$ with the middle again being smaller than p^k. Now note that we can rearrange this string $w = uv_1 \ldots v_kxy_k \ldots y_1z$ as $uzv_1 \ldots v_kxy_k \ldots y_1$ such that $v_1 \ldots v_kxy_k \ldots y_1 < p^k$. Now clearly if we remove $v_1 \ldots v_ky_k \ldots y_1$ we don't remove more than p^k strings (note strong inducison here gives us the first 1 to p^k strings so we can assume that the previous p^k from a given n is true, right now we're showing for the case n assuming the previous are true.). Now note that the string uxz can be derived by taking $S \to^* uAz \to^* uxz$ and hence $\Psi(uxz)$ is in $\Psi(B'_U {C'_U}^*)$ as well. So we can write this as ab for which $a \in B'_U$ and $b \in {C'_U}^*$, now note that b is such that it is a collection of strings in C'_U such that for each string we can go from $A \to^* xAy$, now clearly the strings we removed $v_1v_2 \ldots v_ky_k \ldots y_1$ are such that there is some non termainla such that for any $1 \leq i \leq k$ we have $A \to^* v_iAy_i$ and hence each of them are in C'_U and hence the collection of the k strings we removed are in C'_U. Using the same idea putting them back in the string $uzx = ab$ we still have $abv_1y_1 \ldots v_ky_k$ (note we're putting them back in a differnet ordering but we're preserving the counts of each i.e the result of Ψ for simplicity we don't arrange in the exact order because it doesn't matter), so because the strings we added are in ${C'_U}^*$ and so is b their concat is in it as well and as a is in B_U we essentially have the string that $w = uv_1 \ldots v_kxy_k \ldots y_1z$ is in $B'_U {C'_U}^*$ as well, completing our inductive step. ∎

1.3 Additional Problems

- Problem 6: Show that L_U is in fact a context-free language.

- Problem 7: Use Parikh's theorem to show that every context-free language over a unary alphabet is regular.

  Proof: Consider the CFL L defined over a unary alphabet. Using Parikh's theorem there is some R such that $\Psi(L) = \Psi(R)$. I.e for any string $w \in L$ there is some string in R such that they are letter equivalent. Now take $w \in L$ let our unary alphabet be some a. Now note that $|w| = n \implies w = a^n$. Now clearly there must exist some string $w' \in R$ such that $\Psi(w') = \Psi(w)$ or in other words $|w'| = n$ as well. But if we define only on a then the only choice is if $w' = a^n = w$. So we just showed for any $w \in L$ we have $w \in R$. Similarly for any string $w \in R$ we can show that if $w = a^n \implies |w| = n$ and if $w' \in L$ such that $|w'| = n$ then we have $w' = a^n = w$ and hence $w \in L$. This gives us $L = R$. ∎

- Problem 8: A set S is linear if for some fixed $a_0, a_1, ..., a_n$, we can write $S = \{a_0 + x_1 a_1 + x_2 a_2 + ... + x_n a_n \mid x_i \in \mathbb{N}\}$. A set is semi-linear if it can be written as the union of linear sets. Show that if L is context-free, $\Psi(L)$ is semi-linear. (Hint: use the fact that $\Psi(L_U) = \Psi(B'_U C'^*_U)$)

- Problem 9: Show that if $\Psi(L)$ is semi-linear, then there is a regular language R such that $\Psi(R) = \Psi(L)$. (Hint: start with the linear case, then use the union property for regular languages.)